Sathwik P. Kallapur

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SUMMARY

Software engineer with 3+ years of experience in backend systems, platform orchestration, and high-performance computing. Proven track record in optimizing API performance (50x speed increase at Latlong) and refactoring mission-critical backend logic. Recently completed Master of IT at Monash University with strong academic performance.

WORK EXPERIENCE

Orchestration Software Engineer April 2026 – Oct 2026
NEP Australia *Melbourne, Australia*
- Orchestrated bare-metal servers on Canonical MaaS by building distributed Go workers for Conductor workflows.
- Engineered automated E2E test pipelines and deployment lifecycles using Robot Framework, Jenkins, and Python.
- Built custom Linux OS images with MaaS Packer and Ansible for production broadcast environments.

Software Development Engineer (SDE) 1 July 2023 – Dec 2023
Medpho (Telimedic Technologies) *Remote, India*
- Developed full-stack features and AI chatbots from scratch using Node.js, Next.js, PostgreSQL, and Dialogflow.

Software Development Engineer (SDE) 1 July 2022 – July 2023
CoffeeBeans *Bengaluru, India*
- Refactored financial calculation logic by replacing floats with a custom type, ensuring perfect precision.
- Optimized backend data handling for e-commerce and AI recommendation engines, increasing throughput.
- Maintained multi-DBMS architectures and implemented complex concurrency patterns in Go.

Software Engineering Intern Sep 2021 – June 2022
Latlong (ONZE Technologies) *Bengaluru, India*
- Migrated API infrastructure from Ruby on Rails to Go, yielding 50x faster execution.
- Built high-performance RESTful APIs for Geocoding and Driving Directions using PostGIS.
- Developed a Geocoding algorithm 2x more accurate than Google Maps API for the local context.

TECHNICAL SKILLS

Languages: Golang, C/C++, Python, Java, Rust, JavaScript, Ruby
Backend: Node.js, Next.js, FastAPI, Flask, REST APIs, Microservices, Concurrency
Data & ML: PostgreSQL, MongoDB, Redis, OracleDB, PostGIS
Infrastructure: Linux, Ansible, Jenkins, Docker, OCI, AWS, Terraform, Git, CI/CD

EDUCATION

Monash University Feb 2024 – Dec 2025
Master of Information Technology (MIT) *Clayton, Australia*
- GPA: 3.31/4.0. Key units: Cloud Computing, Advanced DBMS, Discrete Optimisation.
- US equivalent GPA: 3.88/4.0. (Calculated via WES iGPA tool)

PES University July 2018 – July 2022
Bachelor of Technology in Electronics and Communications *Bengaluru, India*
- Major GPA: 8.16/10. Minor in Computer Science (GPA: 8.25/10).
- Focused on Data Structures, Image Processing, Computer Networks, Generic Programming, and Operating Systems.

PROJECTS

Racing Line Mapper | *Golang, Reinforcement Learning, OpenCV* 2025
- Reinforcement Learning system mapping optimal racing lines using MDP and image-processed track skeletons.
- Developed a custom arcade physics engine and transformed raw track maps into semantic Frenet frames.

SafeTrek | *FastAPI, Vue3, PostGIS, Docker* 2025
- Interactive road safety analysis platform using geographic visualization to support urban policy decisions.
- Implemented complex geospatial queries in PostGIS and containerized the entire stack for deployment.

Generic Red-Black Trees | *C++ Template Programming* 2021
- STL-compatible RBT implementation with bidirectional iterators and custom predicate support.

INTERESTS

High Performance Software Design, Game Development, Physics Engines, Sim Racing, Rock Music, and Mindfulness.